

ANNEX E: LIFECYCLE GUIDANCE

**Version 5.0
30 September 2016**

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(U) TBD/TBR

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(U) REVISION SUMMARY

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<i>Version</i>	<i>Revision Date</i>	<i>Revision Summary</i>
1.0	12/31/2015	Initial Release
2.0	2/29/2016	Version2
3.0	4/30/2016	Version 3
4.0	6/30/2016	Version 4
4.1	8/4/2016	Updated classification/portion marking based on review to push down to SBU
5.0	9/30/2016	Update to Sect 3.3 Exploration and Experimentation

(U) GLOSSARY OF ABBREVIATIONS AND ACRONYMS

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<i>Acronym</i>	<i>Description</i>
A2SI	Agency Agility Strategic Initiative
ACC	Advanced Campaign Cell
AoA	Analysis of Alternatives
ART	Agile Release Train
ATDD	Acceptance-Test Driven Development
BDD	Behavior-Driven Development
C2S	Commercial Cloud Services
CD	Continuous Delivery
CIA	Central Intelligence Agency
CIO-T	Office of Chief Information Officer and IT Services Directorate
CONOPS	Concept of Operations
CORE	Consolidated Operations Research Explorer
CP	Capability Project
CPM	Capability Project Manager
DoD	Department of Defense
GIAT	Geospatial Intelligence Advancement Testbed
GS	GEOINT Services
GSM	GEOINT Solutions Marketplace
IAE	Integrated Analytic Environment
IC	Intelligence Community
IT	Information Technology
JIE	Joint Intelligence Environment
MIB	Mission Integration Board
MSA	Major System Acquisition
MUG	Mission Users Group
NGA	National Geospatial-Intelligence Agency
NSG	National System for Geospatial Intelligence
O&S	Operations and Support
PIB	Portfolio Integration Board
PL	Project Lead
PM	Project / Program Manager
PMI	Project Management Institute
POM	Program Objective Memorandum
POR	Program of Record
QA	Quality Assurance
SAFe	Scaled Agile Framework
SE&I	Systems Engineering and Integration
SEP	Systems Engineering Plan
VPC	Virtual Private Cloud
WIP	Work in Process

1.0 (U) EXECUTIVE SUMMARY

(U) This Annex describes the cloud-enabled transformational changes to the NGA Information Technology (IT) lifecycle and provides guidance to both the organization and development engineers.

(U) This NGA Lifecycle Guidance is to document and assist in the evolution of the processes, practices, culture and tools that define the governance, execution and sustainment of NGA's IT operational baseline. This lifecycle guidance identifies the Scaled Agile Framework (SAFe®) 4.0 framework as the integration mechanism for NGA transformation and conformance and is discussed in Section 2.3.

(U) Lifecycle Guidance, Annex E of NGA Cloud Guidance, is for enterprise organizations and industry partners as related to the governance and execution of the NGA IT lifecycle. This guidance is expected to continue until the NGA Systems Engineering Plan (SEP) is released late 2016.

(U) A summary of this guidance is:

- Follow the CIO-T Operating Model
- Use DevOps
- Utilize Dew Point for Exploration and Experimentation activities

2.0 (U) INTRODUCTION

(U) NGA seeks to become an open agile enterprise to be more responsive to its customer needs. Openness initiatives include the migration to a horizontal service-based architecture and the institutionalization of new processes, such as GEOINT Solutions Marketplace (GSM). Agility initiatives include the adoption of the SAFe, cloud migration and the introduction of new contract constructs such as buying development capacity through fixed-priced sprints.

(U) This Lifecycle Guidance v5.0 describes the enterprise-level approach to open and agile IT lifecycle management at NGA. It includes principles, processes, governance and policies that enable NGA to transform into the supplier of choice for GEOINT services and solutions to the IC and DoD.

(U) The overall goal of this Guidance document is to promote, inform and steward the transformation of the NGA enterprise through the embrace of new IT processes and practices to include an agile requirement process, an increased delivery cadence, DevOps framework and an emerging exploration and experimentation environment called Dew Point.

2.1 (U) IT Transformational Overview

(U) This lifecycle guidance is part of the Open/Agile enterprise transformation guidance set, enumerated in [Figure 1](#) below, includes planning guidance (cloud strategy, Tier 1 schedule, etc.) and design guidance (IaaS/PaaS/SaaS, new development stacks, etc.), and leverages and aligns to other NGA transformational initiatives such as the Agency Agility Strategic Initiative (A2SI), the GSM, FastTrak efforts, and the CIO-T Operating Model.

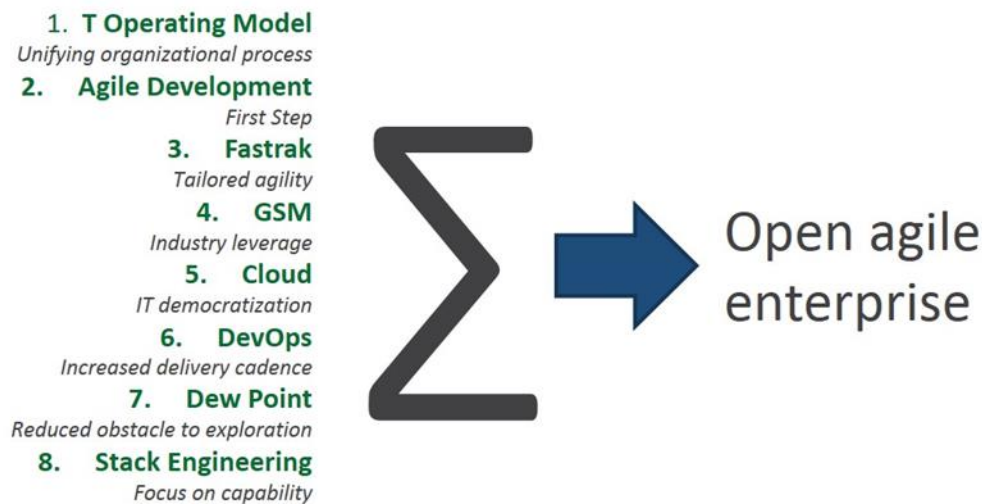


Figure 1: (U) NGA's Transformation Initiatives

(U) NGA is currently pursuing four concurrent operational transformations: mission, cloud, agile, and governance. These transformations are discussed here to provide context and alignment to the lifecycle.

(U) **Mission** – Mission transformation relates to increasing NGA's value proposition in light of the need to rapidly respond to the user community's geospatial needs and apply the growing GEOINT commercial marketplace. GEOINT services have become ubiquitous in the government and commercial communities. Users have access to orders of magnitude more GEOINT data than ever and the commercial market place is moving towards providing more through small/micro-satellites. These small/micro-satellites provide readily deployable multi-spectral, persistent commercial-based GEOINT services. It is expected that given the financial scale of the commercial play, the GEOINT market will be driven to provide GEOINT analysis and faster response times. The scope of commercial ventures has a profound impact on NGA's value proposition and concept of operations.

(U) **Cloud** – Cloud is both an integrating fabric and an IT fabric. As an integrating fabric, the cloud enables community-integrated missions as demonstrated by the IC Advanced Campaign Cell (ACC). It also enables the realization of processing and analytical concepts as those being pursued by the DNI Mission Users Group (MUG) campaigns and the restructuring of NSA's ongoing Major System Acquisition (MSA) efforts.

(U) Cloud intrinsically makes NGA more agile by providing an on demand, elastic IT capability and by mitigating the up-front IT investment barriers. This fiscal flexibility increases the set of industry players that can be leveraged and provides for a larger set of new ideas that can be piloted without much cost and/or risk through environments such as Dew Point. Dew Point is a cloud-based data science solution whose architecture is configurable (e.g., scripts) such that a project can adopt or adapt it, then deploy it to support their own services and workloads within this architecture.

(U) **Agile** – The agile movement increases the flow of capability delivery where capabilities are delivered faster and in a more continuous manner. The degree of enterprise agility can be categorized in three levels:

1. Agile software development: This level relates to the application of agile techniques (e.g., scrum, extreme programming, SAFe team level, etc.) to the software development process. NGA has

adopted variants of the SAFe as its agile development approach and has piloted it in various programs and efforts.

2. Agile Systems Engineering and Integration (SE&I): This level relates to the application of scalable agile techniques such as SAFe (i.e., program and portfolio levels) to integrate programs and efforts. NGA has demonstrated this level in the Integrated Analytic Environment (IAE) effort where various activities were integrated to deliver a set of capabilities in an agile way. NGA has pioneered some agile contract techniques to augment PORs and create new buying patterns allowing the purchase of services and capabilities in a more granular fashion.
3. Agile Enterprise: This level relates to the institutionalization of agile practices in the enterprise. This may require new buy patterns, new contracts constructs, material and non-material practice changes, human capital alignments, etc. NGA is effecting this transformation through the Agency Agility Strategic Initiative (A2SI).

(U) **Governance** – NGA is undergoing change in its governance structure. That NGA is driving the enterprise through the Portfolios is a-work-in-progress. The portfolios' roadmaps and schedules focus on planning and needs. These areas are being worked in unison to formulate the coherent financial and programmatic plan for the delivery of the desired capabilities

2.2 (U) CIO-T Operating Model

(U) **Purpose**: NGA's CIO-T Operating Model provides a fast, flexible, synchronized process for CIO-T to deliver solutions. The Model works end-to-end to integrate strategy, solution, execution, and operations support. It brings in the best solutions available. It helps CIO-T engage users throughout the process to be more responsive to their needs and ultimately deliver the capabilities needed to achieve their mission.

(U) **Overview**: The CIO-T Operating Model is a framework that defines how CIO-T will deliver capabilities and outcomes in a repeatable, systematic way based on predictable planning horizons. The Operating Model aligns to the CIO-T Guiding Principles and focuses on collaboration at every level, continuous engagement with both users and the IT marketplace, and prioritized speed of delivery organized around capability projects that span concept to disposal. Four situational issues drove the CIO-T transformation:

1. Organizational Disconnection: Need to integrate and manage the flow from strategy to delivery. Processes need to be universally followed.
2. Timeliness: Customers say that it takes too long to deliver capabilities. CIO-T was taking too much time to build and implement solutions.
3. Custom Builds: Rather than making use of others' advances (reuse), CIO-T spent an undue amount of time and resources providing custom solutions.
4. Inaccurate POM Requests: Some requirements are pursued with insufficient research. Technologies need to be properly vetted. POM requests can be off-target, directing resources to the wrong activities, and may not lead to sustainable solutions.

(U) **Lifecycle Phases**: The CIO-T Operating Model lifecycle is organized in phases from the time a capability is requested to the time a capability is deployed in the NSG, as depicted in [Figure 2](#):

- Integrated Strategic Planning (*Orange* quadrant)
- Integrated Solution Planning (*Green* quadrant)
- Integrated Execution (*Blue* quadrant)
- Integrated IT Operations (*Brown* quadrant)

(U) All phases are broken down into activities. Activities are broken down into subordinate steps performed by a single function. Functions are aligned to CIO-T organizations. There is a functional overlay that describes the steps performed within an activity.

(U) The CIO-T Operating Model is a framework to visualize and understand how work collaboratively happens. The Model's phases are defined as follows:

- Strategic Guidance is the high-level strategic direction received from Agency and Community leadership. It takes the form of future visions, strategic plans, Agency-level priorities, and Department of Defense and Intelligence Community mandates.
- Integrated Strategic Planning translates Guidance into projects through concepts of operations, business cases, and the understanding of market trends.
- Integrated Solution Planning gives projects detail by defining capabilities, planning acquisition, and integration into the architecture.
- Integrated Execution delivers solutions and capabilities by executing on project plans and managing the change to the enterprise.
- Integrated Support to Operations provides ongoing maintenance, support, and retirement of solutions.

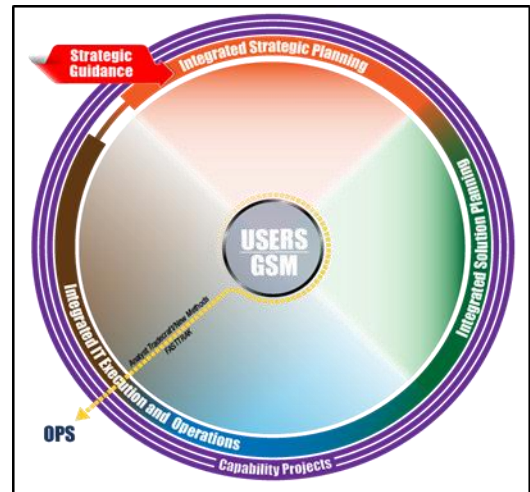


Figure 2: (U) CIO-T Operating Model Phases

(U) [Appendix A](#) – CIO-T Operating Model Lifecycle Phases provides a more detailed definition to the lifecycle phases of the CIO-T Operating Model and a capability projects discussion.

(U) End-to-End Projects: provide mission capabilities to perform either a technology's added-value or satisfy a timely mission user need.

- FastTrak is a rapid deployment process intended to bring capabilities and applications quickly to users. FastTrak doesn't circumvent the Operating Model, it is intended to support the rapid insertion of analyst tradecraft and new methods. Once acquired through FastTrak, capabilities can be scaled up and supported by the enterprise by using the Model.
- Capability Projects and Operational Mission Threads continually bring consequential capabilities to our customers. They extend through all of the phases and provide continual feedback from users. The capability projects drive end-to-end integration of the Operating Model.

(U) **Core Elements:** The CIO-T Operating Model is built around Users and the GEOINT Solutions Marketplace (GSM). The Users are NGA's customers and their mission needs. NGA's customers require a broad range of services, including the delivery of geo-referenced imagery and data, Foundation GEOINT products and services, and intelligence analysis. Increasingly, many customers operate on dramatically reduced timelines, driving a need for persistence - collection at the speed of the observable - from multiple sources. The GSM is a collaborative and transparent online ecosystem that bridges the gap between NGA and solution providers in industry and academia. The core elements effect on the Operating Model are as follows:

- Users are at the center of the Operating Model. Users provide feedback, give insight into evolving needs, and prioritize capabilities. User engagement is critical to continuous improvement. Users are what drive the Model, keeping CIO-T focused on the mission.
- GEOINT Solutions Marketplace provides broad capability and solution options from industry, academia, and other government agencies. Here, best practices and innovations come together in a sandbox environment to share insights and influence new solutions.

(U) The Operating Model is an end-to-end process, not serial. It contains on-ramps and off ramps that will differ depending on the needs of each project. Projects don't have to begin from scratch to run through the Model. For example, existing background documentation from prior projects can be reused.

(U) The Model guides the accurate and rapid delivery of capabilities. It synchronizes strategy and development to provide greater efficiency, better use of resources, and unifies our support to customers. It helps CIO-T make POM requests that are on-target and meet specific, well-understood needs. It focuses on the user and embraces innovation from the marketplace. It embodies the flexibility to manage changing priorities and resource constraints.

(U) In its entirety the Operating Model is intended to balance structure and agility, standards and responsiveness, and scalability and speed. It gets us away from the tyranny of the urgent, and helps us invest in important resources in the future. It gets us back to the GEOINT basics, back to what we're good at doing.

2.3 (U) Integrating Framework

(U) NGA has adopted the Scaled Agile Framework (SAFe). This framework as shown in [Figure 3](#) will serve as the integrating fabric to align the Operating Model, DevOps and many process and cultural changes. The following sections map to the SAFe framework and show how the topics integrate via the NGA-accepted SAFe 4.0 framework.

(U) **SAFe V4.0:** The SAFe 4.0 framework addresses mostly the material aspects of the enterprise and has four levels of abstraction mapped to NGA's efforts as follows:

1. SAFe Portfolio level: The SAFe portfolio level corresponds with the NGA enterprise where requirements are baselined. Enterprise epics are built at this level.
2. SAFe Value Stream level: The SAFe Value Stream level maps to NGA portfolios constructs. Where groups of efforts providing a given value to the enterprise are managed (Analytic, content, etc.).
3. SAFe Program level: The SAFe Program level maps to NGA programs.
4. SAFe Team level: The SAFe Team level maps to the development effort supporting a program.

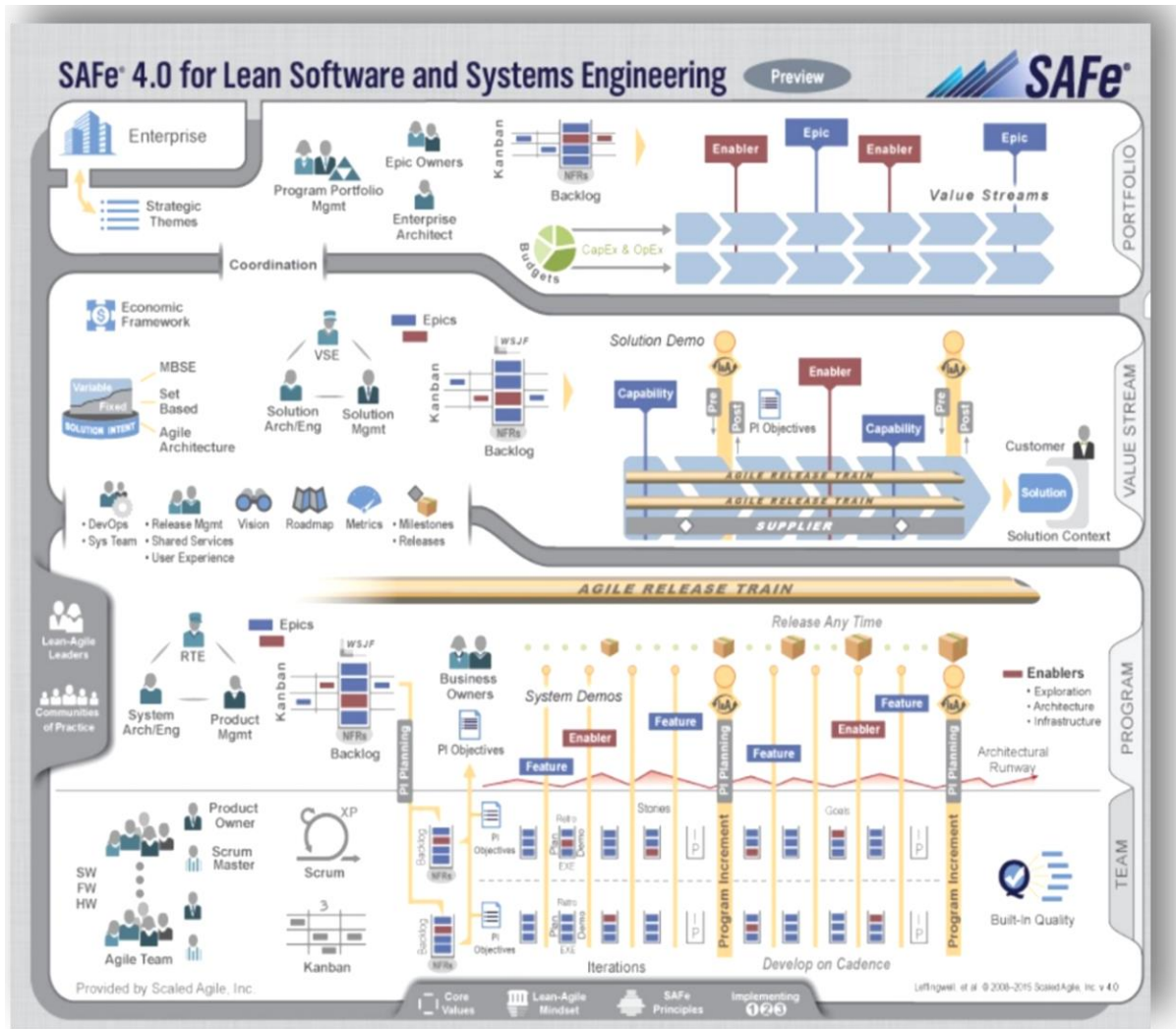


Figure 3: (U) Scaled Agile Framework 4.0

(U) **CIO-T Mapping:** The CIO-T model maps to the SAFe 4.0 framework as shown in Figure 4 below in the following lifecycle phases:

1. **Integrated Strategic Planning:** Integrated Strategic planning maps to the SAFe 4.0 portfolio level where strategic guidance is decomposed and aligned into value stream epics and capabilities.
2. **Integrated Solution Planning:** Integrated solution planning maps to the SAFe 4.0 value stream level where value stream epics and capabilities are decomposed into program epics and features and assigned to agile release trains.
3. **Integrated Execution and Operations (DevOps):** Integrated Execution and Operations maps to SAFe 4.0 program and team levels where features are decomposed into user stories. The CIO-T model follows SAFe such that both integrated execution development and operations teams support the DevOps structure.

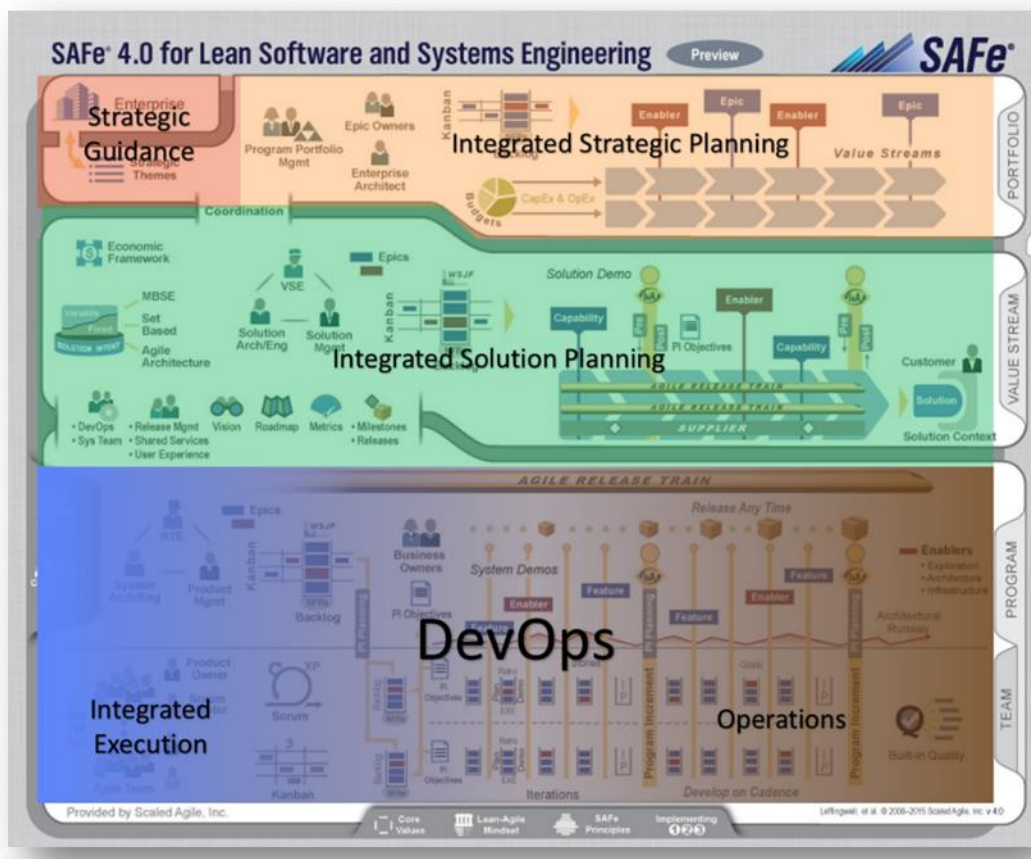


Figure 4: (U) Scaled Agile Framework 4.0 – CIO-T Mapping

(U) DevOps Mapping: As mentioned previously, DevOps practices are implemented at multiple levels of a large enterprise such as NGA. In the SAFe model, program-level DevOps practices are implemented at the SAFe Program level by Agile Release Trains. Agency-level DevOps practices are implemented at the SAFe Portfolio and Value Stream levels.

(U) The NGA DevOps strategy recognizes that a successful enterprise DevOps implementation requires an approach that focuses on changes at three different levels:

1. (U) **Program**

- DevOps at the program level is concerned with implementing practices and tools to streamline the continuous delivery (CD) pipeline for a single development program to increase the flow of mission capabilities to users.

(U) Outcome: Define the common NGA DevOps patterns and control points.

2. (U) **Agency**

- DevOps adoption at the Agency level addresses institutionalizing DevOps principles, culture, roles and responsibilities, and processes across the agency, but more specifically focuses on integration activities between programs, release trains and value streams.

(U) Outcome: Expand the DevOps process and principles to the enterprise level to ensure integration and realization of solution/function delivery in support of a value stream. Institute DevOps as a desired enterprise development pattern including refactoring of enterprise functions such as requirements, configuration management, etc.

3. (U) **Intelligence Community**

- DevOps at the community level relates to NGA’s implementation of DevOps practices and tools in the context of its role as both a producer and consumer of technology-enabled services within the cloud-enabled mission ecosystem of the Intelligence, Defense, and Civil government communities.

(U) Outcome: Leverage Community development resources (e.g., DevOps Factory, DI2E, GEOINT Services, etc.) and rapidly develop interoperable and reusable services within the Community mission ecosystem.

(U) Enterprise related Lifecycle Processes

(U) NGA, as all large enterprises, has a lifecycle business process for various aspects of its business operations. These agency lifecycle processes (Strategic, Acquisition, POAM, etc.) must align to the IC and DoD enterprise lifecycle constructs. The NGA engineering lifecycle must align to two key enterprise processes, which are the Annual Strategic Process and the Acquisition Lifecycle, as shown in [Figure 5 and 6](#) below. These processes are not the subject of this discussion, they are included for completeness and reflect the enterprise connection to the topics discussed herein.

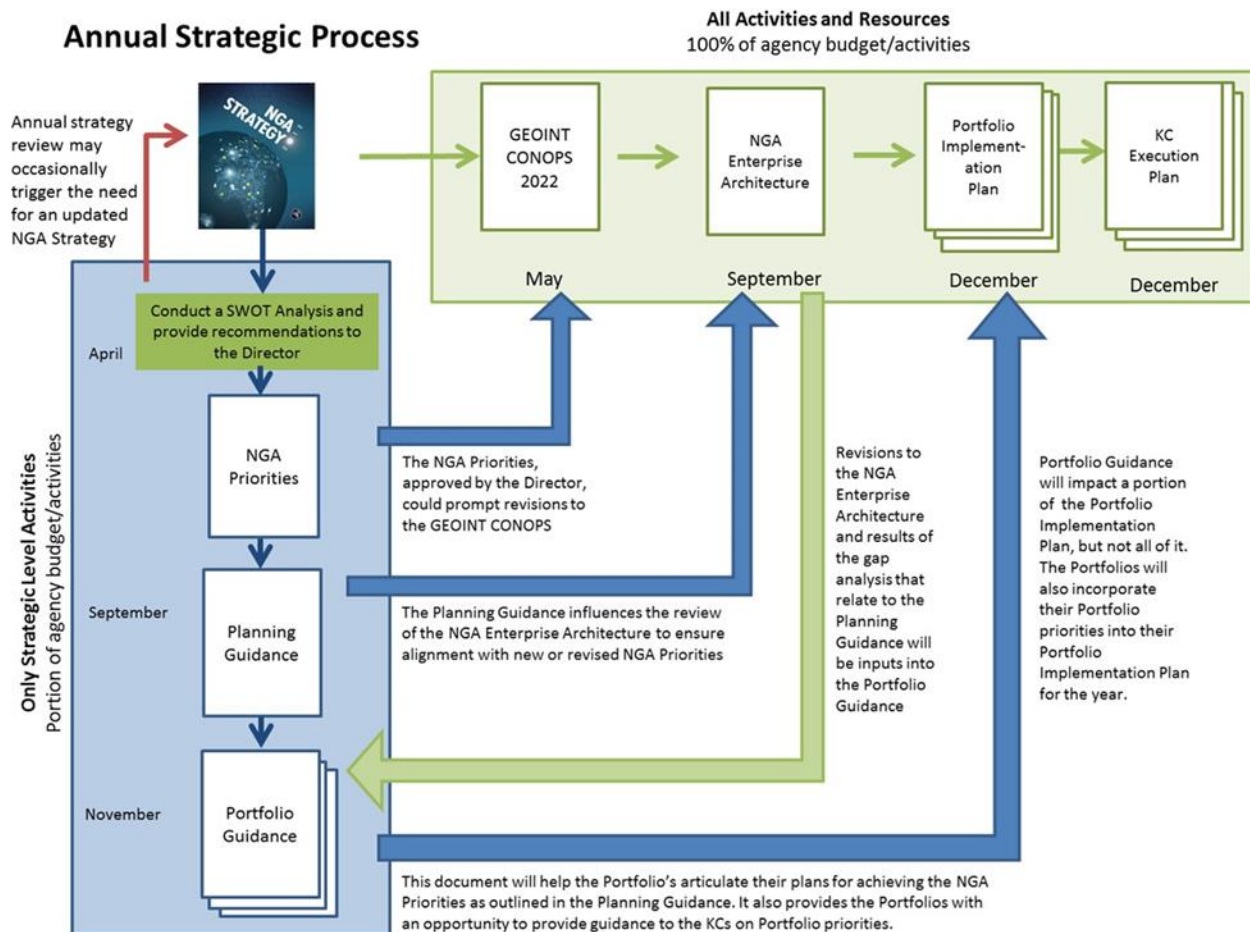


Figure 5: (U) Annual Strategic Process

(U) The strategic process seeks to realize the NGA GEOINT 2022 CONOPS¹ through the continued evolution of the NGA priorities, planning guidance and portfolio guidance. The CIO-T model aligns and enables the enterprise strategic process mostly through the evolution of the baselines managed at the SAFe portfolio and program levels.

(U) The NGA acquisition process is being adjusted to increase its agility and ease of use. Examples include the purchase of Agile sprints through fixed price vehicles and increased use of broad instruments such as GSA mechanisms. In addition, a buy-versus-build mindset is being promulgated across the enterprise with initiatives such as the GEOINT Marketplace and Pathfinder efforts. In addition, acquisition is facilitating instruments that support more agile-accommodating contract vehicles.

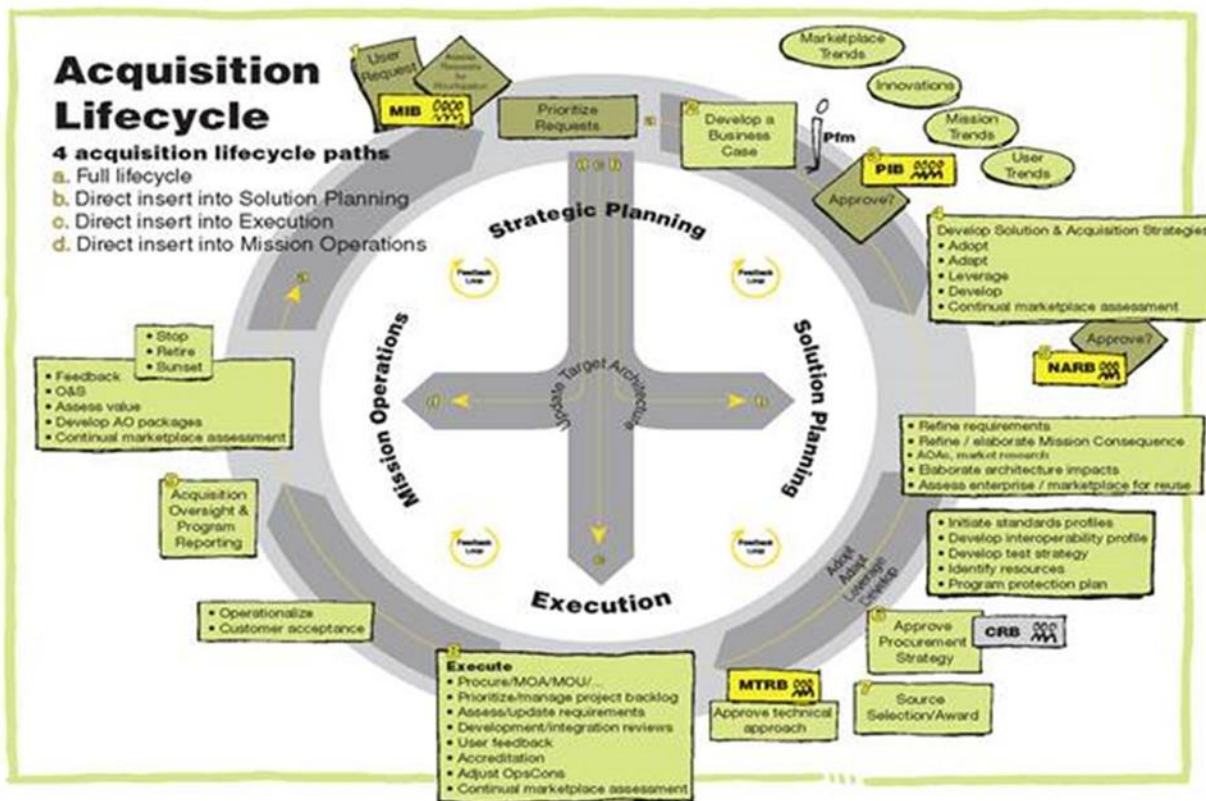


Figure 6: (U) Acquisition Lifecycle

(U) The acquisition lifecycle shown above in [Figure 6](#) directly aligns to the CIO-T Operating Model, and supports increased agency agility and leverages industry technology.

2.4 (U) Operational Principles

- (U) Implement the CIO-T Operations Model
- (U) Implement SAFe 4.0 for Lean Software and Systems Engineering as the integrating framework that allows the organization to use the Operations Model and DevOps
- (U) Operate within the overarching lifecycles of the IC and DoD

3.0 (U) KEY TRANSFORMATIONAL LIFECYCLE CHANGES

¹ The [NGA GEOINT 2022 CONOPS](#) is the description of a broad application to the whole GEOINT mission, vice solving a specific mission need.

3.1 (U) Requirement Process Transformation

(U) One of the most affected areas of the engineering lifecycle is the requirement process. While there are expected changes at the enterprise and portfolio requirement process level this document will focus on the transformation happening at the program level.

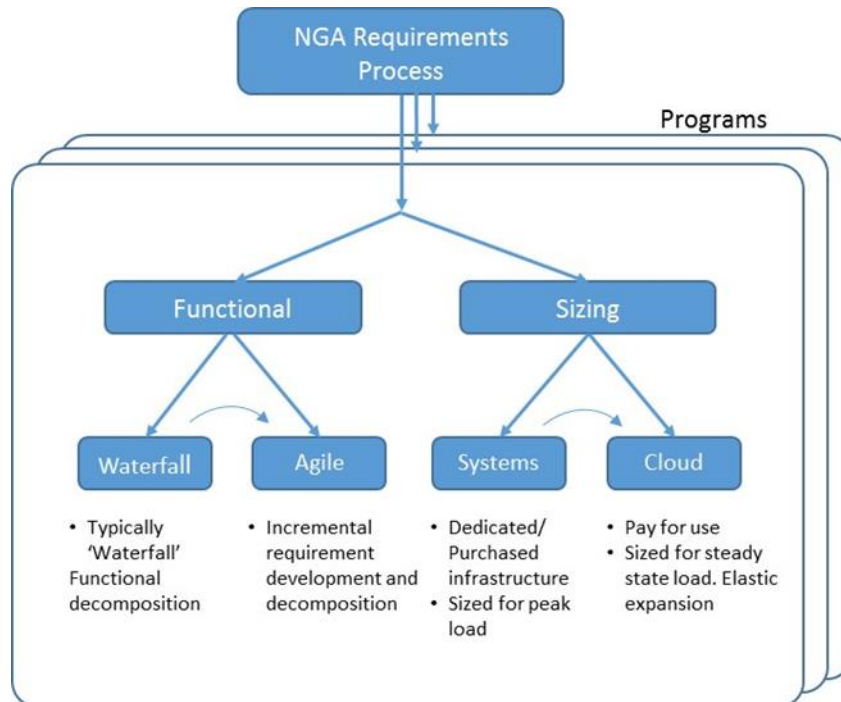


Figure 7: (U) Functional Decomposition and Infrastructure Sizing

(U) In general, the program requirement process is focused on two objectives: 1) functional decomposition and 2) infrastructure sizing. Both of these areas are being transformed, as shown above in [Figure 7](#) from Waterfall to Agile, a significant departure in construct.

(U) Functional decomposition seeks to decompose the high level program requirements into specific testable requirements. In the waterfall legacy lifecycle this is usually accomplished all at once through hierarchical decomposition analysis that usually lasts 3 to 6 months for most programs. Once this is accomplished and reviewed, the requirement baseline is established for the program. As shown below in [Figure 8](#), the agile process is based on the SAFe framework that uses an interactive and incremental requirements process. For most efforts NGA is moving from a waterfall to agile lifecycles; and, as such from design, develop, test, operate to an incremental design/develop/test/operate.

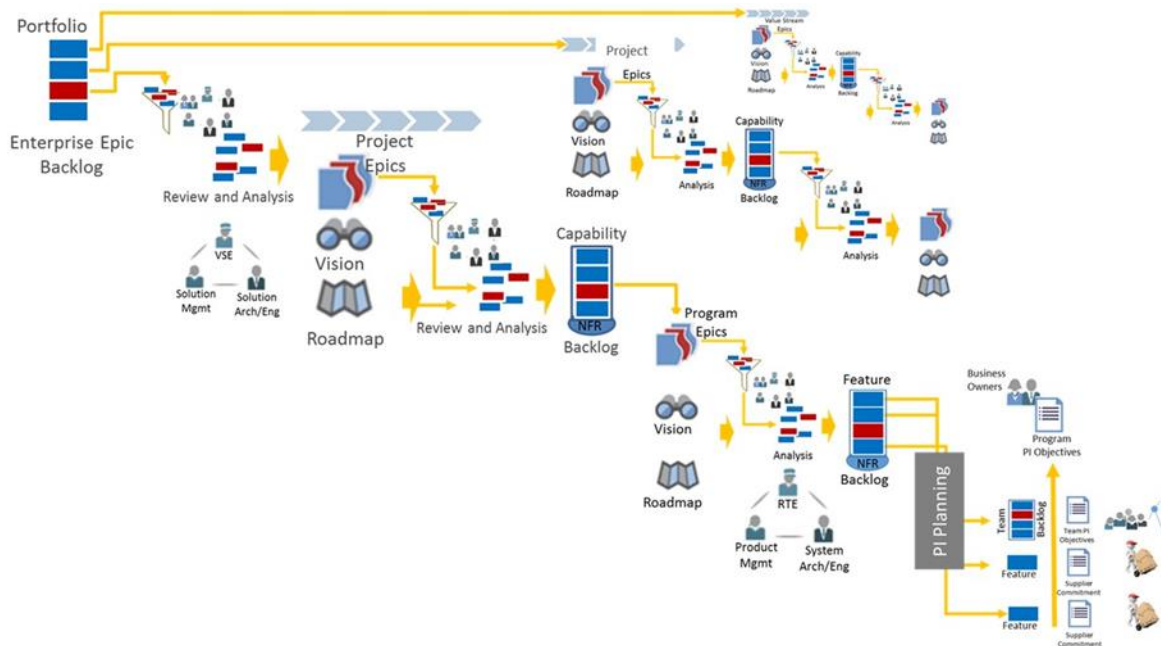


Figure 8: (U) Interactive and Incremental Requirements Process

(U) Infrastructure sizing is the process focused on sizing both the software and hardware resources so they can be purchased and setup in the selected data center. Historically, at NGA this process takes about 9 to 12 months and involves multiple organizations with infrastructure design, build, and operations functions. The cloud pay-for-use model has changed this process. While the programs are required to do an initial cloud resource estimation, programs can instantiate development resources in the cloud from day one. In addition, the use of the cloud elastic computing can reduce the cost incurred. Through the use of elastic computing, the need for more capacity can be achieved on demand and the only pay-for-use factor greatly reduces the actual operating costs.

3.1.1 (U) Operational Principles

- (U) Transform the requirements process from waterfall to agile and from system to cloud
- (U) Focus requirements on two objectives: 1) functional decomposition and 2) infrastructure sizing

3.2 (U) DevOps

(U) DevOps is an evolution of agile development, where small batches of code are developed, integrated, tested, and deployed in frequent, short cycles utilizing software development best practices, and cultural and organizational barriers between applications developers and IT operations personnel are reduced to increase the pace and efficiency of deploying software updates to users. This is in contrast to traditional development approaches which revolved around large batches of requirements that resulted in sequential, large batch handoffs between requirements definition, software design, software test and verification and deployment to operations, ultimately leading to significant delays, high rates of defects, and high program risk. The lower complexity of small changes minimizes the risk to operations and in so balances the scales between the ever changing needs of the development environment with the perpetual stability needed in an operational environment.

(U) DevOps is a significant paradigm shift which impacts culture, processes, people, and tools at multiple levels of the enterprise. Guidance version 3 is primarily focused on program and team level practices and tools. Developers are the primary users of Continuous Delivery tools and practices. Government Product Managers, Product Owners, and Release Managers will be responsible and accountable for ensuring that prescribed DevOps guidance is followed. Users will receive the benefits of rapid capability deliveries by working closely with development teams, test teams, security engineering teams, and IT operations teams.

(U) Annex E provides the guidance for how NGA will adopt and execute a DevOps methodology to accelerate the "time to market" of ideas into operational software capabilities in support of national security. Adoption of DevOps at NGA will increase its business agility. In executing DevOps, NGA will realize a transformation of software development and IT operations. Embracing this construct will yield savings in time and funds for development, efficiencies for personnel performing in the agile construct, and customer satisfaction with earlier deployment of higher quality software and services.

(U) DevOps adoption is about more than just adopting new technical tools. In order to realize the improvements in IT Operations performance and GEOINT mission performance that NGA seeks, programs must adopt the DevOps culture, embrace lean and agile management practices, and implement Continuous Integration (CI), Test Automation, and Continuous Delivery (CD) tools and practices in their development environments.

(U) This Annex outlines an agile and decentralized approach to governance at the lowest level that enables development, security engineering, and IT operations teams to collaborate together efficiently and minimize delays and barriers to deploying working software frequently. It also outlines NGA's guidance for provisioning, adopting, and/or maintaining the CI/CD/Test Automation tools that enable DevOps, and provides a measurement framework for assessing the maturity of programs implementing DevOps practices to assist them in their continuous improvement efforts and to identify and resolve enterprise impediments that inhibit program agility and value delivery.

(U) Adoption of the DevOps construct has essentially begun with NGA's use of the SAFe methodology. DevOps adoption within NGA is the 3-level construct (i.e., Program, Agency, and Community) noted in [Section 2.3](#). Implementation will occur as NGA moves to the cloud and will continue to mature into normal engineering and business processes by the time NGA completes the move to the cloud in December 2017.

(U) [Figure 9](#) below shows the relationships of the 3 levels from capability development at the program level, through the Agency Portfolio level, to the Community level where NGA is both a member and a contributor of integrated capabilities. This figure is described in detail in the following section.

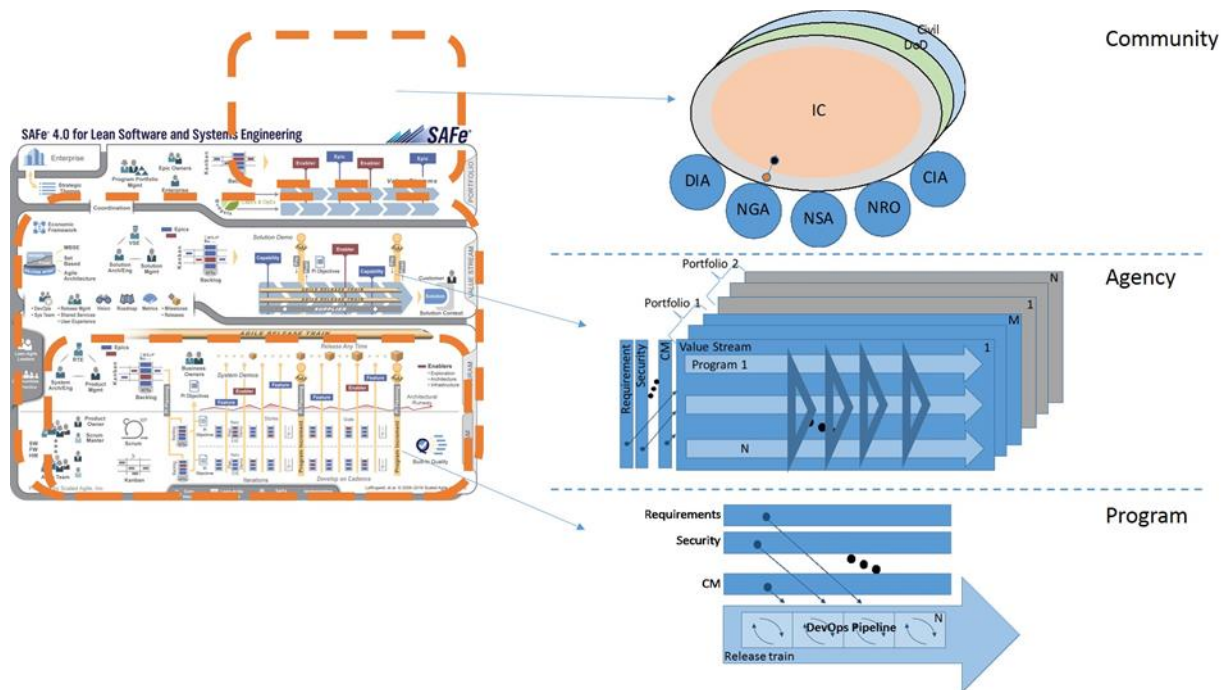


Figure 9: (U) Dev Ops at Three Levels

(U) Why Adopt DevOps?

(U) NGA can accelerate the time required to deliver new capabilities or update existing capabilities by embracing DevOps practices. Whether developing new, original software code or integrating COTS or GOTS software applications or services, adoption of DevOps practices and tools can help NGA software adoption and delivery activities to increase their efficiency, delivery pace, quality, and fit for purpose, while reducing delays, defects, and program risk. As an integral part of the overall CIO-T transformation, DevOps adoption requires commitment from across the NGA enterprise to changes that span people, process, and technologies. Core tenets of the DevOps paradigm include the following:

- **Culture:** People and process, organizational alignment and responsibilities
- **Automation:** Technology and tools to accelerate key tasks and reduce errors
- **Lean:** Utilizing lean, flow-based management practices, including decentralizing decision-making and visualizing and managing Work In Progress (WIP)
- **Measurement:** Situational awareness and process insight through metrics
- **Sharing:** Establishing feedback loops and enabling collaboration and reuse

(U) The shift in to DevOps is not without cost and some level of risk in the transition. It requires a culture change from NGA's approach to engineering. Waterfall development has given way to agile. Culture change is more than engineering, development, test, and deployment changes occurring within CIO-T; it includes all programs, all portfolios, and extends into the IC for integration of all programs in support of national defense. Key improvements that will be realized include:

- New mission capabilities arrive more quickly at less cost with far fewer defects due to the continuous testing, integration and delivery
- Increased efficiency of fielding new capabilities and reduce waste
- Increase mission utility and mission user satisfaction through fast user feedback
- Multi-INT mission capability integration in the IC ITE ecosystem

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“Getting to a DevOps agile development probably is the primary driver” for transforming the NGA Information Technology Services Directorate.

Doug McGovern, D/CIO-T, Signal Magazine, April 1, 2016

(U) Implementation Strategy

(U) It is NGA’s goal to fully embrace DevOps by October 2016 and to contribute fully to the implementation and evolution of DevOps in the intelligence community. NGA’s implementation approach is an enterprise-wide roll-out and requires a broader implementation strategy that includes the 3 levels (Program, Agency, and Community) previously discussed. The implementation will be initiated by learning with a few programs already applying DevOps principles and practices. Then, expanding implementation of DevOps principles and behaviors to existing programs as appropriate; and, to all new acquisition program starts in the future. Measures of progress will be identified early and continuously captured as NGA matures DevOps implementation. These measures will demonstrate where improvements and efficiencies are being realized, as well as drive out refinement to the DevOps implementation. The following sections will introduce and discuss each of the Program, Agency and Community levels. A “crawl, walk, run” approach will be identified for each level, along with supporting tasks/actions that are required to support NGA’s approach.

3.2.1 (U) Program Level

(U) Definition

(U) DevOps at the program level as shown in [Figure 10](#) is concerned with implementing practices and tools to streamline the continuous delivery (CD) pipeline for a single development program. NGA programs will utilize a CD pipeline that rapidly and continuously promotes new software code through four stages. Program implementation requires support from key enterprise functions, such as requirements management, configuration management, IT security, architecture, quality assurance/IV&V, and others. In the DevOps paradigm, some of this support may be provided via dedicated personnel or cross-functional teams assigned directly to the ART, while some will be provided by standing enterprise functional teams and working groups operating in an integrating role. An ART will normally have a single CD pipeline, but in certain cases, such as in the case of a program with teams supporting multiple products with different or independent technical architectures, an ART may have multiple CD pipelines.

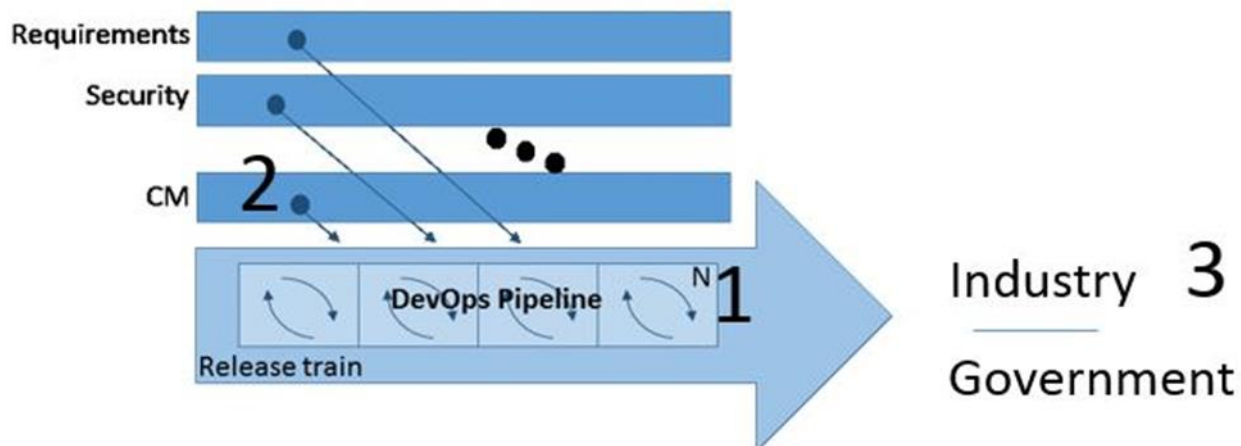


Figure 10: (U) DevOps at the Program Level

(U) For the purpose of this discussion, the term “program” is used here in a broad, general sense, as defined by the Project Management Institute (PMI)². The intent is to refer to a cohesive, standing organization of software development and sustainment teams, without regard for the contractual or financial arrangements for procuring these teams.

(U) It is further stipulated for the purpose of this discussion that the default organizational model for a software program is the SAFe Agile Release Train (ART). Implementing DevOps requires first achieving a measure of proficiency and maturity in implementing team-level agile practices such as Scrum, XP, or Kanban, and at a larger scale, requires application of multi-team agile patterns for scaling these practices such as the SAFe ART. Programs that are not implementing a SAFe ART model or a similar organizational pattern for scaling multi-team agile development are unlikely to realize the benefits of implementing DevOps practices.

(U) Intent

(U) The intent is to achieve three key goals:

1. Normalize a core set of common DevOps processes, tools, and behaviors across NGA programs
2. Institutionalize execution of key enterprise functions in support of programs implementing DevOps practices
3. Specify and manage the minimum set of control points necessary to ensure streamlined governance, enable rapid, decentralized decision-making, and realize the benefits of DevOps tools and practices

(U) Tasks

(U) The implementation strategy for DevOps at the program level is focused on two major activities: 1) collaborative engagement with a subset of NGA software development activities currently implementing DevOps practices to elicit lessons learned, effective practices, and pilot proposed guidance, and 2) providing guidance to enable upcoming new-start programs to operate from day one using NGA’s normalized program DevOps tools and approach.

1. Establish default program CI/CD technical architecture (aka “DevOps toolchain”):
 - a. Establish a [list of trusted/vetted tools](#) and identify procurement guidance for CI/CD/Test Automation infrastructure tools³
 - i. Maximize economies of scale for procurement of CI/CD/Test Automation software tool licenses, including opportunities to leverage Community managed infrastructure
 - b. Develop guidance for tool configuration
 - c. Approach for continuous monitoring – application performance/state of health, security performance, and business analytics
2. Develop and implement a program-level measurement framework for programs implementing DevOps tools and practices:
 - a. Survey pilot/pathfinder programs to identify measures and metrics currently being collected and reported

² “A program is a group of related projects, subprograms, and program activities managed in a coordinated way to obtain benefits not available from managing them individually.” *PMI PMBoK, fifth edition, 2013*

³ DevOps tools listing is maintained in the NGA Cloud Information Portal ‘s [Cloud Services Catalog](#).

- b. Develop and iteratively refine a program-level maturity model focused on program and team practices, and collaborate with DevOps pathfinder programs to conduct regular self-assessments concurrent with program Inspect & Adapt rhythms
 - c. Develop and track IT performance measures (e.g., lead time, deployment time, deployment frequency, deployment success, test automation, defect density, time to correct defects, MTTR, system stability, etc.)
 - d. Develop and track mission/business value delivery metrics (user net promoter scores, mission performance metrics)
- 3. Develop program-level guidance:
 - a. Define and manage a set of program-level control points to enable streamlined governance and decentralized decision making
 - b. Define the template for support to ARTs by key enterprise functions: configuration management, IT security, QA/IV&V, requirements management, architecture, etc.
 - c. Define key government and industry roles and responsibilities in program execution under a DevOps paradigm
 - d. Systematically identify, prioritize, track, and resolve enterprise impediments to program-level continuous delivery
 - e. Recommend contractual language for program DevOps
- 4. Organizational development activities. While the most significant organizational development activities are at the agency level, a few tasks are required to drive the program-level changes in practices, tools, and culture.
 - a. Define and implement a strategy to evolve program culture.
 - i. Leader development. Shifting to a DevOps culture requires a systematic approach to training NGA senior and mid-level leaders responsible for program execution in lean flow-based management practices
 - ii. Workforce training. Government and contractor personnel responsible for implementing DevOps tools and practices require training in these practices

(U) Roadmap

(U) **Crawl:** In the “crawl” stage, program level DevOps implementation is focused on successfully implementing CI/CD practices on two or more Agile Release Trains operating independently, measuring key practices and outcomes, identifying effective practices and desired control points, and implementing DevOps “First Way” practices associated with managing flow. In the “crawl” stage pilot/pathfinder programs will demonstrate the following behaviors and outcomes:

- Programs track and report measures against a common measurement framework, addressing metrics related to process maturity, process execution, software quality, system performance, user satisfaction, and mission performance. Programs conduct periodic self-assessments against an NGA DevOps maturity model
- The technical architecture for the program CI/CD pipelines (“DevOps toolchain”) are documented
- Programs have implemented basic Test-Driven Development practices
- Programs have implemented a production-equivalent staging environment and are deploying working code to staging at least once every iteration/sprint

- Programs produce potentially shippable software every iteration/sprint
- Programs visualize and manage work In Progress (WIP) at the ART level using the SAFe program level Kanban system
- Programs implement ART-level Inspect & Adapt rhythms, share key DevOps lessons learned, implement process improvement tasks in their program backlogs, and resolve or elevate impediments related to DevOps to the DevOps Roundtable
- Programs hold regular engagements with business/mission owners to solicit actionable feedback on feature priorities, program roadmap, and demonstrations of working software products

(U) **Walk:** In the “walk” stage, NGA will shift focus toward normalizing the DevOps approach across programs, synchronizing development and delivery across two or more ARTs within a value stream, progressing the maturity of program DevOps practices to at least maturity level 3, and implementing DevOps “Second Way” practices to optimize feedback. In the “walk” stage pilot/pathfinder programs will demonstrate the “crawl” level behaviors plus the following additional behaviors and outcomes:

- A standard technical architecture for program CI/CD pipelines (“DevOps toolchain”) is defined and available to all NGA programs
- Programs have demonstrated growth in overall maturity to an average of Level 3 across all measured activities in the NGA DevOps maturity model through a systematic application of the measurement framework and consistent process improvement through the ART-level Inspect & adapt rhythms
- Programs have implemented Acceptance-Test Driven Development (ATDD) or Behavior-Driven Development (BDD) practices
- Programs produce potentially shippable software every iteration/sprint and are able to deploy to an NGA production environment on demand
- Program cadences are aligned and synchronized with other ARTs in the same value stream. Programs participate in solution demonstrations from a value-stream-level production-equivalent staging environment
- Program backlogs, roadmaps, and release plans are aligned and synchronized with the value stream-level solution backlog, roadmap, and release plan
- Programs are implementing NGA DevOps lifecycle guidance with governance according to a limited set of well-defined control points
- All NGA new-start programs are implementing DevOps tools and practices according to NGA DevOps lifecycle guidance

(U) **Run:** In the “run” stage, NGA will focus on increasing the number of programs implementing DevOps tools and practices, advancing the maturity of program DevOps practices to levels 4 or 5, and implementing DevOps “Third Way” practices to drive experimentation and innovation to improve GEOINT mission outcomes and advance the tradecraft. In the “run” stage programs implementing DevOps practices will demonstrate the “walk” level behaviors plus the following additional behaviors and outcomes:

- Programs habitually execute their Inspect & Adapt and continuous process improvement rhythms and have demonstrated growth in overall maturity to an average of Level 4 or 5 across all measured activities in the NGA DevOps maturity model
- Programs have fully automated the creation of development, test, and staging environments

- Programs have fully automated the deployment process
- Programs are implementing hypothesis-test driven development with frequent user feedback, including automated, real-time business intelligence
- NGA continues to adapt and evolve DevOps lifecycle guidance based on lessons learned, feedback from programs and value streams, and emerging industry practices and innovations

3.2.1.1 (U) Operational Principles

- Implement the continuous delivery (CD) pipeline for two or more SAFe Agile Release Trains (ART)
- Specify and manage a minimum set of control points
- Implement a program-level measurement framework

3.2.2 (U) Agency Level

(U) The Agency level addresses DevOps principles, roles and responsibilities, and process across the agency, but more specifically focuses on integration activities between programs, release trains and value streams. The Agency level, as shown in [Figure 11](#) below, will include an enterprise source code repository that supports the integration of code across programs and release trains as necessary to support end-to-end testing and ensure readiness for delivery to operations.

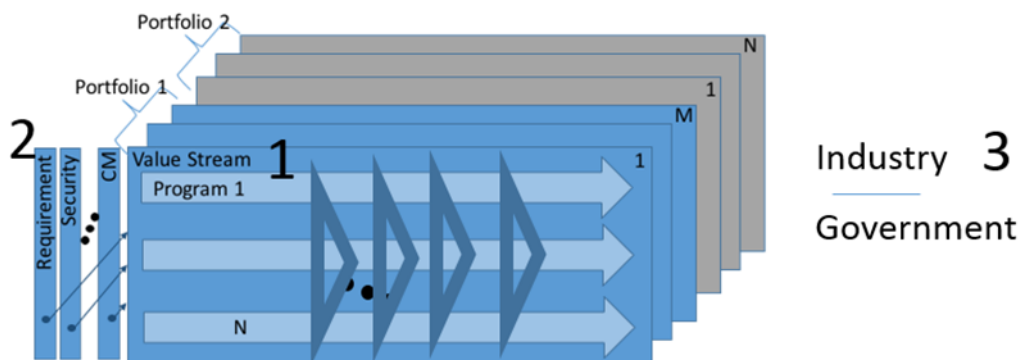


Figure 11: (U) DevOps at the Agency Level

(U) Intent

(U) The intent is to expand the DevOps process and principles to the enterprise level to ensure integration and realization of solution/function delivery in support of a value stream.

(U) Tasks

(U) Implementation of the Agency level for DevOps includes a shift from a program/system focus to an ART focus and implementation. This shift will push alignment and integration at the Agency level. Below are a list of key tasks to start the implementation approach. Tasks 1 and 2 are provided to include the broader agency level context and are assumed to be in place or will be implemented by agile processes at the agency level. For the purposes of this DevOps at the Agency level implementation, it is assumed that a value stream (e.g., ASC) has been defined and is aligned to the Analysis portfolio.

1. Value Stream Definition – A value stream (or investment system/solution) is the series of steps that an Enterprise uses to provide a continuous flow of value to the customer (i.e., from the concept idea through solution delivery). A value stream will be identified and funded by one or more NGA portfolios.
2. Value Stream Management – The Agency level will include the establishment of Solution Manager(s) that will provide the overall leadership and ensure coordination and integration of activities supporting the Value Stream. Additionally, the Solution Manager will work with other Solution Managers to ensure coordination across value streams as needed.
3. DevOps at the Agency Level – The foundational construct of DevOps at the agency level is realized by the establishment and implementation of the Agile Release Train. Specifically, an ART that supports the ASC value stream. DevOps is in integral part of the ART and therefore, in order to move forward with DevOps implementation is critical that an ART is stood up and working as an integrated activity. The ART is an integrated activity with teams from the Program level working with System Architects, Business Owners, System Engineers, Security and Test and the Product Manager to understand and implement the value stream vision and related Features. DevOps at the Agency level will support enterprise level integration, testing, security, and release approval/readiness.

(U) The ART is the organizational construct that will realize a value stream. A value stream may be supported by one or many ARTs, depending on the overall size of the effort.

- a. Determine approach to train alignment
- b. Identify the roles/responsibilities for ART management and execution
- c. Identify the monitoring and measuring characteristics of an ART. How will we know if we are successfully delivering value (i.e., solution/functionality to the end user/customer)?
- d. Identify and define the shared staging environment. What will it look like and how it will operate at the Agency level?
- e. Identify the control points where code posted to lower level (program level) repository(ies) are pushed up to the Agency level repository
- f. What are the control points at the Agency level?
- g. Determine how will visualize and manage WIP at the agency level
- h. What tools are needed to support the Agency level DevOps?
- i. Identify key leaders in TI/TA/TO/Others that need DevOps/Agile/SAFe training
4. Monitoring/Measuring
 - a. Measuring performance of the value stream. What will we measure?
 - b. How will we monitor progress and determine “definition of done” and success?
5. Industry/Government Partnership
 - a. What are the roles/responsibilities and interactions between the Government and Industry?
 - b. How will the Government communicate to Industry? What information will the Government communicate to Industry?
 - c. What support/functions does the Government need Industry to provide?

(U) Roadmap

(U) **Crawl:** Establishing value stream understanding and communication with NGA portfolios. Resolving impediments to Programs that are “crawling”. Preparing agency level processes and roles/responsibilities to support “walk” at the Program level. Educating/training key TI/TA/TO/other leaders in DevOps/Agile, SAFe.

(U) **Walk:** A minimum of one value stream has been established. Conducting solution demonstrations from a shared staging environment. Work in Process (WIP) is managed at the value stream level. Value Stream roles are established. Government and Contractor personnel are fully trained/qualified.

(U) **Run:** All NGA portfolios implementing at least one value stream. Robust business intelligence is collected and analyzed and informing refinement and modifications (continuous learning). Real-time performance monitoring is in place.

3.2.2.1 (U) Operational Principles

- (U) Implement an Agile Release Train (ART) construct at the Agency level.

3.2.3 (U) Community Level

(U) DevOps at the community level relates to the DevOps activities and interactions NGA has with the Intelligence, Defense and Civil government communities as shown in [Figure 12](#) below. In these interactions NGA has four roles: provider and user of mission capabilities and provider and user of DevOps capabilities.

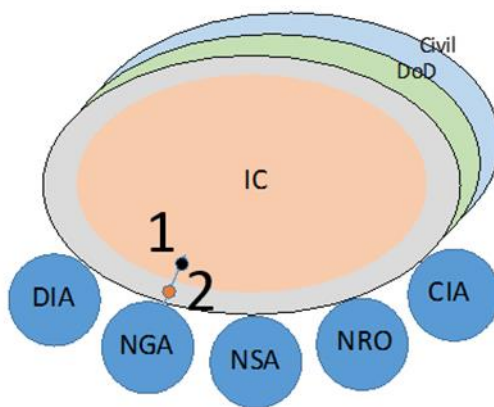


Figure 12: (U) DevOps at the Community Level

(U) Intent

(U) The IC ITE and Joint Intelligence Environment (JIE) environments have elevated the mission context from an agency to a community level. These environments support the development and execution of community wide integrated missions such as DNI MUG missions and MSA redesign. In this context agencies provide and use each other’s services and functions instead of just providing intelligence products.

(U) It is expected that the current agency level DevOps environments such as DevOps factory, Consolidated Operation Research Explorer (CORE) and GEOINT Services (GS) will be integrated and/or leveraged with each other to provide a DevOps platform (Gray band in picture above) for emerging community integrated missions and doing in common what is commonly done it. This community level DevOps capability will evolve through time and it is expected to normalize to a hybrid implementation

model such as DIA’s CORE DevOps platform. CORE leverages the CIA’s DevOps Factory capability and concurrently instantiates some of the DevOps features locally. This approach leverages common functions while allowing agency specific needs to be addressed.

(U) Tasks

(U) It is expected that the Community DevOps Roundtable will serve as the steward of the DevOps community level capability. In this role the round table will serve as a focal point to orchestrate and guide the discussion of how best to leverage and integrate each agency DevOps capability. In support of this the round table should:

1. Define, provide oversight and promulgate DevOps at the community level

The Community DevOps Roundtable should be established as an entity sponsored by the DNI and providing the stewardship of DevOps at the community level. The NGA DevOps Charter could be used as model to be scaled up.

The DevOps Roundtable should perform the following functions:

- Be a self-governed team identifying common ground for DevOps
- Provide guidance on the DevOps approach
- Engage on requirements, configuration management, security and business processes impacting NGA’s DevOps adoption
- Provide recommendations on managing the DevOps approach, tools, and control gates
- Provide lessons-learned guidance and implementation templates to programs
- Address the DevOps challenges that affect NGA programs
- Engage and collaborate with other IC agency DevOps advisory groups

(U) The DevOps Roundtable serves the DNI as a technical component providing oversight of the IC DevOps activities. The Roundtable’s principal responsibilities are to:

- a. Advise and support the DNI in implementing DevOps procedures, processes and best practices.
 - b. Maintain transparency of DevOps implementation activities.
 - c. Communicate implementation lessons-learned from migrating programs and ongoing DevOps activities.
 - d. Represents the IC in other communities’ DevOps related forums
2. Foster visibility, and transparency of DevOps developments across the community

The roundtable should serve as the focal point for DevOps exchanges across the community and should instill collaboration and a community level mindset across its DevOps participants. The DevOps Roundtable meetings should focus on the implementation and configuration management concerns confronting DevOps employment. On a recurring and open way and in order to maintain visibility and transparency the roundtable should:

- a. Review actions
- b. Review status/progress of DevOps activities
- c. Address DevOps issues/challenges and assign responsible parties for specific follow-up actions

- d. Review DevOps best practices, lessons-learned, and emerging tools, as appropriate
- 3. Steward the piloting of alternative solutions supporting community and agency DevOps needs.

The roundtable should provide guidance in the piloting of DevOps solution. Through the establishment of AoAs and through a visible and transparent review mechanisms alternatives the roundtable should drive a re-use and common capabilities mindset where community needs come and are leveraged first before agency specific implementation are pursued.

- 4. Enable Inter- and intra-community DevOps synergy.

The roundtable should engage and collaborate with other IC agency DevOps advisory groups.

(U) Roadmap

(U) The standup, implementation and execution of DevOps at the community level is expected to take a couple of years. An evolutionary path is suggested with each evolution involving both the community DevOps, and each Agency DevOps capabilities. Based on a hybrid implementation approach such as CORE, the suggested roadmap baselines a community offering from which the specific agency implementation is then developed. It is expected that one agency would be selected to pilot the next addition for the community baseline. Once successfully demonstrated this capability will be baseline at the community level and “rain-down” on the supporting agencies. It should be noted that several concurrent capabilities could be in pilot mode across the IC.

(U) The roundtable should establish a community DevOps roadmap and the associated piloting roadmap and assignments.

3.2.3.1 (U) Guidance to the Organization

- (U) Participate in Community DevOps Roundtable

3.3 (U) Exploration and Experimentation

(U) Systems engineering at NGA has been, and to a large extent still is, tied to the traditional systems engineering “V,” which reflects waterfall development and expensive systems development over a period of years which yielded an end result that may or may not have achieved the success envisioned by the customer. Requirements for a given system are aggregated and integrated into a desired system. Engineering occurs over a period of years and deployment occurs after extensive testing and refinement. Investment in terms of cost, time, and resources is significant. Risk of failure after years of development is high.

(U) With the advent of the cloud this style of engineering must embrace change. The cloud’s on-demand utilities and the waterfall engineering approach are incompatible. Time for development needs to be in terms of days or hours, not months or years. Security is not an end-state, but is incorporated consistently with software builds. Storage and compute resources are commodities, not required in the development of capabilities.

(U) NGA seeks to create a climate that encourages experimentation and risk-taking. Why? This will allow the agency to securely develop new capabilities, test them, and produce desired products without risk of impacting systems. Failure need not result in collateral damage and collapse of a program at great cost in funds, time, and personnel.

(U) Rapid experimentation and evaluation of new ideas with a ‘fail early, fail often,’ and fix before proceeding approach is desired. Experimentation risk should be based on the idea being tested, not on the effort or cost of setting up the testing. The failure of an idea should not result in the loss of significant money or time beyond that invested in developing the idea itself. This includes (historically) buying hardware and/or (currently) provisioning real or virtual hardware through a request process, then obtaining security approvals and accreditation of the test configuration; progressing from the Geospatial Intelligence Advancement Testbed (GIAT), to a pervasive exploration capability.

(U) The desired solution to avoid the pitfalls of traditional systems engineering must:

- Leverage just-in-time development
- Promote high quantity/volume of experimentation
- Minimize time-to-test and cost-to-test
- Provide access to NGA data and algorithms
- Provide for the ability to adapt quickly to rapidly changing customer needs
- Allow for secure continuous integration, continuous improvement, and continuous deployment

(U) In order to realize this rapid experimentation, NGA must transition from a set of infrequently changing “monoliths” to a loosely coupled suite of “microservices”. Under a microservice model, each service can evolve largely independent of the other services as long as the exposed interfaces maintain compatibility. This allows a much higher pace of change across the environment, while allowing consumers to adopt new features at their own pace.

(U) An essential part of realizing this model is having well-documented application programming interfaces (APIs) that detail the specific methods and parameters supported by each service. Industry has embraced an “OpenAPI” specification that can be used to describe, in machine-readable format, the details of an API’s implementation. By treating these APIs as contracts between different parts of the IT organization, we facilitate specialization and reduce the cross-organizational coordination required when making changes.

(U) Another potential innovation accelerator is the increasing availability of “managed services” within the cloud ecosystem. On JWICS, C2S has several managed service providers working to make their capabilities available “as a Service”. These service providers will receive some level of “authority to operate” (ATO) from their sponsor, allowing customers to benefit from this accreditation. The on-demand nature of these “pre-baked” services allows customers to quickly reach a point where they are delivering mission value vice expending time and energy on the mundane tasks of configuring and hardening servers. NGA is working to provide a set of platform services that reduce the work necessary to build, deploy, and operate geospatial applications.

(U) A common thread across many of these efforts is the use of containerization technology to simplify the transition of IT capabilities through a lifecycle. Much like microservices helps simplify functional interactions, containerization helps to simplify the temporal interactions across an application’s lifecycle. In some cases the developer may explicitly manage container configurations, while in other cases the platform leverages containers “under the hood” to deploy and run custom code. By creating applications that perform service discovery at runtime rather than utilizing static configurations, the resulting application can be deployed in multiple environments unmodified. This yields a number of significant benefits, including:

- Parity across development, test, and production environments

- Rapid scaling and recovery from hardware failures
- Simplifying application upgrades and updates
- Establishing clear lines of responsibility between parts of the IT organization

(U) Another critical enabler for exploration is version control. “Versioning all the things” makes testing and deployment a highly repeatable process, one that can be automated to catch issues before they reach the production environment, and one that allows (in the worst case scenario) redeployment of a “known good” baseline. Versioning is a cornerstone of modern build pipelines, enabling vast efficiency gains by encapsulating the application build and deployment process as code.

(U) Any discussion of exploration would be incomplete without acknowledgement of the role that security plays across the lifecycle. Exploration involves change, and change is typically viewed with great skepticism by security professionals. Successful exploration must therefore consider where and how (within the technology “stack”) different security-relevant decisions are being made, and focusing change control efforts on these components. By understanding this “what, where, and how” of security, developers can quickly insert new functionality that leverages existing code or services vice spending time on non-value-added-but-required efforts.

3.3.1 (U) Operational Principles

- (U) When choosing between competing implementation approaches, give precedence to the approach that can best support future changes without requiring modifications by external consumers their service.
- (U) Wherever possible, deliver capabilities as one/more HTTP-based API’s. Adhere to the [open-api.org](https://openapi.org) specification in documenting the supported methods, request parameters, and responses.
- (U) Avoid implementations that require “direct” database or filesystem access across program boundaries.
- (U) Maintain awareness of the “managed service providers” available within the cloud environment and the specific categories of problems they seek to address.
- (U) Leverage container or platform-as-a-service technology to simplify delivery of IT capabilities.
- (U) Use software version control and a centralized repository (e.g. Gitlab) for managing both code and documentation.
- (U) Assess new features or bug-fixes early in the development process for potential security relevance.

APPENDIX A – (U) CIO-T OPERATING MODEL LIFECYCLE PHASES

(U) Appendix A provides a description of the CIO-T Operating Model’s lifecycle phases and activities.⁴ Table A-1 identifies the activity, lead performer and objective of each activity by phase in the lifecycle.

Table A-1: (U) CIO-T Operating Model Activities by Lifecycle Phase

Integrated Strategic Planning			
ID #	Activity	Lead	Objective
1	Update Capability Roadmap	EA	Schedule, POM budget, PIB approval
2	POM Budget Build	IT FM	CBBJ
3	Update CIO-T Business Plan	GPO/TCP	Execute Business Plan
4a/b	Synthesize Enterprise Requirements	XFO/OGM	MIB approval/RAMP review
5	Build Business Case	CPM	PIB approval
6	Define Capability	CPM	Capability Roadmap/Program Build
7	Establish Capability Project	CPM	NARB approval/update
Integrated Solution Planning			
8	Define Project Plan	PL	Technical Approach (Epic)
9	Perform AoAs	PL	Technical Approach (Epic)
10	Add Technical Approach to Project Epic	PL	Support 11, 12, and 13
11	Create Program Epic(s)	PL	Execution Plan
12	Estimate Resources	PL	Execution Plan/CRB approval
13	Update Project Plan	PL	CCB approval/Source selection
Integrated Execution			
14a	Project Execution Coordination	PL	Plan and Integrate Release(s)
14b	Program Execution Management	PM	Development and reporting
15a	Plan and Integrate Release(s)	PL	Integrate and deploy
15b	Develop/Contract	PM	Acceptance/Ops transition
16a	Integrate and Deploy	PL	Operate and sustain
16b	Project Acceptance	PL	DevOps/OAR
Integrated Support to Operations			
17a	Project Operate and Sustain	PL	Evolve and update
17b	Program Operate and Sustain	PM	Reporting and oversight
17c	Program Operate and Sustain	Operations	Reporting and oversight
18	Hardware Recap and Disposal	Operations	Refresh/Retire
19	System Monitoring and Reporting	Operations	Identify/Resolve Problems
20	System Retirement	Operations	Strategic Planning

A-1 (U) Strategic Guidance & Integrated Strategic Planning

(U) Strategic Guidance is the high-level strategic direction received from Agency and Community leadership. Strategic direction comes from future visions, strategic plans, Agency-level priorities, and Department of Defense and Intelligence Community mandates. It’s here that the strategic intent of Agency leadership is orchestrated and factored into our Strategic Planning.

⁴ CIOT OP Model Process Flow Guide -- SAFe 4.0 Overlay (v3.1) defines the activities for each lifecycle phase.

(U) Integrated Strategic Planning is the phase to analyze our business intelligence and assess the degree to which success criteria is identified. Current market trends and GEOINT offerings to fit mission needs are evaluated. The GEOINT problem to be solved is defined and explored.

(U) In this phase, the Enterprise Information strategy, the IT strategy, and the GEOINT data strategy are developed. It is essential that resources are available to satisfy current and future NGA mission priorities. Architecture planning is conducted to incorporate the information technology capabilities that will support the Agency vision. IT business and operational resources are aligned under the direction of the strategic guidance. Here, governance by the Mission Integration Board, the Portfolio Integration Board, and the NGA Architecture Review Board focuses on strategic decisions and resource optimization. This phase ensures accountability and compliance with NGA's global IT enterprise.

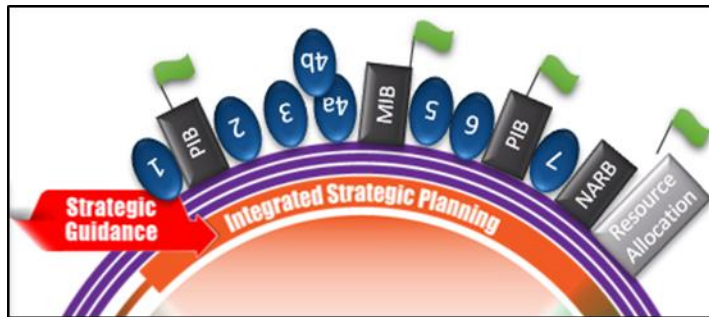


Figure A.1: (U) Integrated Strategic Planning Phase

(U) The four major processes in this phase are:

1. Synthesize Enterprise Requirements: is where triage requests, conflate requirements, and identify groupings of projects, programs, and tasks are identified. The key steps are: 1) to collect requirements from stakeholders, 2) identify program/project groupings, and 3) perform request triage. (ref.: activities 4a/b)
2. Building a Business Plan/Case: completes the due diligence necessary to support the decision to proceed with developing or acquiring the capabilities to support the mission need. (ref.: activities 3 and 5)
3. Define Capabilities: includes naming the capability team, developing the kick-off brief, and developing the enterprise epics. Enterprise epics include success criteria, alignment to DOTMLPF-P, assessment against enterprise and marketplace reuse opportunities, and a technology approach. POM budget builds and CBBJ support are long-term programmatic responsibilities. (ref.: activity 6)
4. Building a Solution Strategy: decomposes enterprise epics into project epics. It identifies project leads and teams. It updates the enterprise roadmap and effectivities. It updates capability and project roadmaps. It provides architecture artifacts. And if necessary, it develops acquisition strategies. (ref.: activities 1, 3 and 7)

(U) Note that the process includes notifications to the chief engineer and the chief architect that the process activities are complete via automated emails (ref.: flags) prior to board action.

(U) Feedback from operations also informs our Integrated Strategic Planning. Lessons learned from having delivered solutions and Operations and Support influence the strategic view. In turn, updates to the enterprise architecture and the capability roadmap provide inputs to adjust planning to adapt and meet customer needs.

A-2 (U) Integrated Solution Planning

(U) Integrated Solution Planning is where CIO-T translates strategy into a solution. The key to this phase is the activities necessary in project planning. This helps understand the users' needs, and whether those needs are nice-to-haves or must-haves. Tradeoffs between capabilities should be completed early, with understanding of industry trends through the GEOINT Solutions Marketplace.

(U) The three major processes in this phase are:

1. Analysis of Alternatives (AoA): is performed to determine if a service or system works well enough to meet mission needs, with input from the mission users. The mission owner determines what's critical, with the goal of leveraging existing capabilities and minimizing the need to build custom code. AoAs must also include architecture and standards constraints to make sure the solutions fit the enterprise. (ref.: activity 9)
2. Technical Approach to Project Epic: develops a technical approach for the project and add it to the epic. Epics and effectivities support one another and provide incremental capability to the users. (ref.: activities 10 and 11)
3. Project Planning: documents the project's technical approach, including the requirements, CONOPS, delivery methodology, resources, and infrastructure. (ref.: activities 8, 12, and 13)

(U) If development is required, the right contract vehicle is determined. The decision to use a new or existing contract normally occurs earlier during the Integrated Strategic Planning phase, but that decision can be refined here as a result of the AoA.

(U) As part of the technical approach, the delivery methodology is determined. Delivery can be done in several ways. Such as, adopt an existing commercial service, leverage another service in NGA or another agency, customize a commercial or agency service with scripts or code, or build custom code tailored to the needs of the users. For build and adapt, preference is given to safe agile when determining the contract vehicle and software methodology.

(U) In the technical approach, the critical design elements, standards, the minimal viable requirements, and nice-to-have requirements, are kept at a high level. Requirements will be further decomposed and prioritized in the Integrated Execution phase of the Operating Model. With the project's technical approach determined, a schedule is built to include dependencies, epics, and effectivities.

(U) All the information is incorporated into a project plan and/or acquisition plan. To finalize the technical approach, resources are estimated, including the funding needed, people, and infrastructure. Enterprise and Architecture requirements are updated based on those plans. Documentation for approval include mission workflows, CONOPs, solution architecture, PMP, training needs, test plans, performance analyses, transition plans, and other non-material changes.

(U) New capabilities may allow for the retirement of older systems and services. A plan to review any new services or systems for obsolescence is developed. Potential retirements are identified in the plan, along with how they will be retired.

(U) If the AoA finds that a commercial or other existing service will meet the minimal needs of the customer, there are two options:

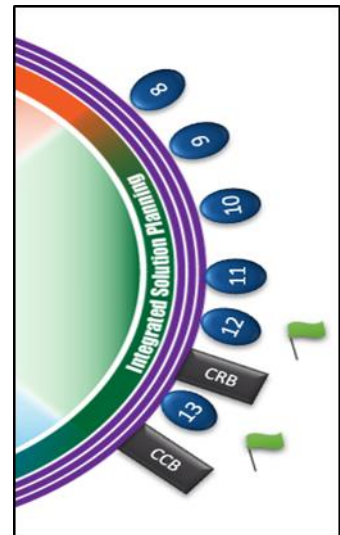


Figure A.2: (U) Integrated Execution Phase

1. Enter the service in the FastTrak process to quickly move it into operations.
2. Initiate a demo or experiment with a representative group of users to determine if the service meets their needs. Based on the outcome of the experiment or demo, either transition the service to the entire enterprise or discontinue its use and update the AoA to find a new solution.

(U) Throughout this phase, it is essential to continually validate progress and decisions with customers and assess what the marketplace can offer, learning together about the art of the possible and fine-tuning knowledge of what will work to achieve mission success.

A-3 (U) Integrated Execution

(U) Integrated Execution delivers solutions and capabilities by executing on project plans and implementing the change within the enterprise. This is all about adoption and delivery of application, infrastructure, and content solutions.

(U) In this phase, the path to sustaining the capabilities is identified; as well as, performing operational and cyber security

readiness reviews. Reviews are effective control points to manage costs, schedules and performance against contracts. Independent verification and validation (IV&V) to include standards and cyber compliance is performed. At the end of this phase, the solution is transitioned into operations.

(U) There are five processes in Integrated Execution:

1. Execution Management: solicits user feedback and prioritizes the backlog based on that feedback. (ref.: activities 14a and 14b)
2. Develop or Contract, and Integrate: means adapting, adopting, or leveraging existing capabilities in addition to the creation of new activities. The Develop or Contract and Integrate process affects and finalizes contracts and contract changes, as applicable. Activities in the process include updating requirements, architectures, interfaces, and schedules. These items are key agenda items in conducting development reviews. (ref.: activities 15a and 15b)
3. Customer Acceptance: includes conducting reviews, testing, and identifying and resolving discrepancies. Acceptance Test and IV&V procedures are also developed. (ref.: activity 16a)
4. Transition to Operations: achieves readiness for the transition, installs the capability into operations, and conducts checkout or tests and transition. (ref.: activity 16b)
5. Compile Authorized Outage packages: are compiled for the purpose of AO scheduling. (ref.: activities 14b and 16a)

(U) Integrated Execution reduces time to deploy new services using technology adoption and agile acquisition processes. And by embedding mission resources throughout the process, customers are more engaged in the success of their solution.



Figure A.3: (U) Integrated Execution Phase

A-4 (U) Integrated Support to Operations

(U) Integrated Support to Operations provides ongoing maintenance, support, and retirement of solutions. In this phase, continuous deployment of projects is undertaken. IT operations and network defense operations are conducted to support external customers. Operations engagement includes assessing IT performance and continually improving IT services to meet mission needs.

(U) Integrated Support to Operations has two major processes:

1. Operations and Support (O&S): conducts operational and mission readiness reviews. It identifies and corrects discrepancies. And it incorporates requirements, revision changes and up-time improvements. (ref.: activities 17a/b/c)
2. Monitoring and Retirement: maintains up-time availability (maintain, update, recap) and executes the disposal of superseded or duplicative services and capabilities. (ref.: activities 18, 19, and 20)

(U) In the CIO-T Ops Model, Integrated Support to Operations provides continuous improvements to GEOINT services in accomplishing the customer's mission. Performance objectives include increasing service availability, security, and accessibility; as well as, reducing the time-to-provision and time-to-respond.

A-5 (U) Capability Projects

(U) Capability Projects (CPs) are the glue that ensure what is needed is what is delivered. They provide seamless continuity of work between each phase of the CIO-T Operating Model. The CP manages feature delivery from external programs. The profile for performing a capability project is highlighted in [Figure A.5](#). Feature Requirements are defined by the CP and passed to external programs. Capabilities are built by external programs and delivered to the CP which integrates and tests externally developed capabilities. The CP provides:

- An Effectivity Roadmap to guide the delivery of capabilities
- Project and mission management to prioritize the backlog
- A Value Stream Engineer to coordinate planning activities
- A System Team to support Feature and end-to-end test activities
- A Systems Engineering team that defines feature capability requirements and architectural direction to drive external project development
- Mission oversight through the mission/business owners to ensure the feature effectiveness are delivering value to the business

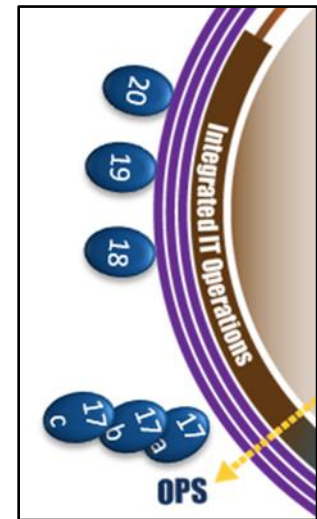


Figure A.4: (U) Integrated Support to Operations Phase

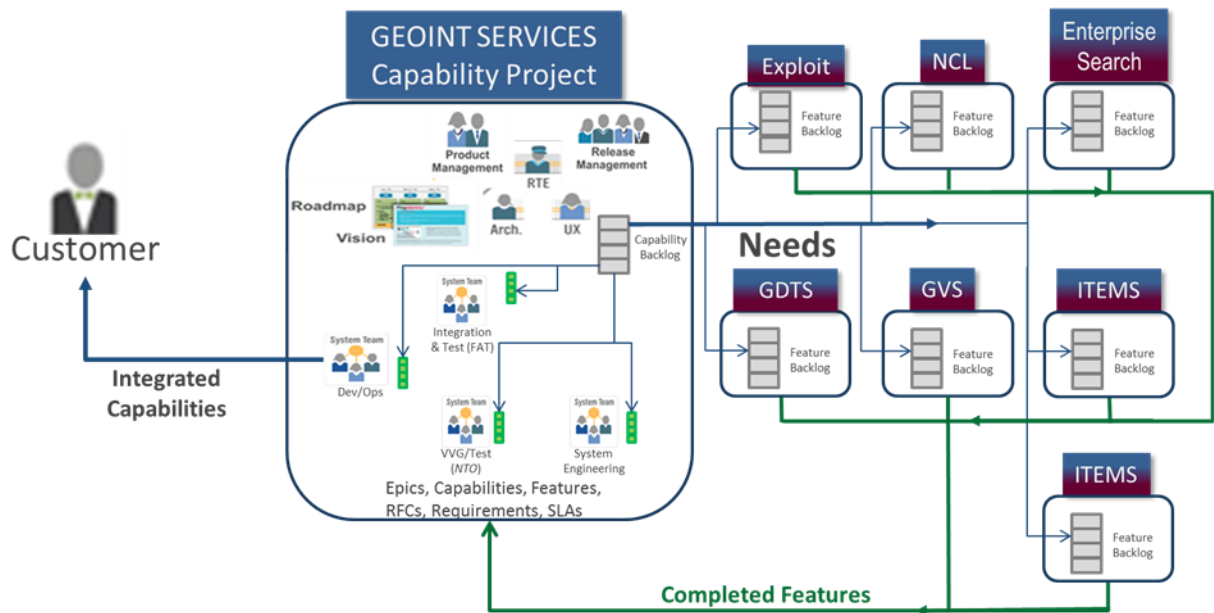


Figure A.5: (U) Capability Project Interaction

(U) As defined in the Operating Model, during the Integrated Strategic Planning phase, projects are identified from the Strategic Plan and allocated to the portfolios. Every project has a project lead, capability project manager (CPM), and mission owner. The mission owner represents the mission; the CPM represented the portfolio, and the project lead (PL)/project manager (PM) represents acquisition. The CPM and the PL/PM work together to develop the business case for the project. This includes researching the marketplace to understand trends and potential solutions that may be ready for limited demonstration or future procurement. The project team makes trade-offs as they develop the business case for the project.

(U) Once written, the Portfolio Integration Board (PIB) approves the business case. Upon approval, the process lead and the mission owner work together to build out the solution strategy. The Mission Integration Board (MIB) approves the solution strategy. The project lead then takes the leadership role as the project moves into the solution planning. The project lead makes decisions following the AoA and delivers challenges to the marketplace through GSM. The mission owner drives the decision of what is considered to be “good enough.” With the requirements baselined, CIO-T creates the solution architecture and updates the target architecture.

(U) The project team ensures that the requirements and schedule is coordinated across the board to include systems engineering: requirements, architecture, training, security, independent verification and validation, standards, and schedules. First and foremost, the project team ensures that the mission needs will be met. Demonstrations or solutions may be selected to go through FastTrak and directly into operations. The appropriate governance boards weigh in on decisions at various levels on various topics as they arise.

(U) The project now moves into execution. Execution includes procurement of software, services, development of service level agreements with other agencies, or execution of a contract to build the service. During this time, the project team ensures that mission needs are understood and addressed by the solution. The DevOps environment helps to ensure these needs are met. The Project Team closes on enterprise needs and keeps the effort within the scope of the business case. If the users request services outside of the scope, the request must go back through the cycle for prioritization.

(U) User representation is included during Test. The project team makes sure that the users' voices are heard while balancing the scope of the project. The Project Team ensures user training and the capability/service transition activities are coordinated. The Project Team continues to support operationalization. When the solution is stable, the team stands down from the project.